National University of Computer and Emerging Sciences
Karachi Campus

Database Systems Lab (CL-2005)

Mid Term Lab Exam

Total Time (mins): **120**

Total Weightage: **25**

Total Marks: **50**

Total Questions: **03**

Paper Type: **B**

Date: Monday, March 30, 2026

Time: 8:30 AM to 10:30 AM

Lab Instructor(s):

Muhammad Khalid

Student Name	Roll No	Section	Student Signature

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Attempt all the questions!

Important Instructions:

- Return the question paper and mention your student ID on it.
- Read each question completely before answering it. There are 3 questions on 3 pages.
- Attempt all the given questions. All questions are carrying different points.
- Cheating in any case will lead to F-GRADE as per university rule.
- In case of any ambiguity, you may make assumptions. But your assumption should not contradict any statement in the question paper.

Question 1: LLO-04 Database Design & Manipulation
[Marks: 14 | Time: 40 Minutes]

A large-scale e-commerce platform operates across multiple cities. The system manages products that are supplied by different suppliers, where the same product may be offered at varying prices by different suppliers.
The platform maintains stock of products at different warehouse locations. Customers can place orders that may contain multiple items. Each order is processed from a specific location, and stock levels are updated accordingly.
Customers may optionally apply discount coupons while placing orders. Each coupon has an expiry date and a limited number of times it can be used. The system must ensure that products are not sold if they are out of stock.

Requirements:

1. Design and create a complete database schema using SQL DDL commands.
2. Your design must include at least five (05) tables with appropriate relationships.
3. Apply suitable constraints where needed.
4. Insert sample data into all tables.
5. Write four (04) SQL DML queries demonstrating:
    - INSERT
    - UPDATE
    - DELETE
    - SELECT

Question 2: LLO-03 Intermediate Queries on the "NorthWind" Database
[Marks: 10 | Time: 20 Minutes]

A company wants to perform a deep analysis of customer purchasing behavior across multiple regions. They are interested in identifying **high-value customers** who contribute significantly to overall revenue, taking into account **order frequency, average order value, and region-specific behavior**.

They define high-value customers as those:

1. Who have placed more than **5 orders in the last 12 months** (Marks: 2)
2. Whose **total purchase amount** is greater than the **average purchase amount of all customers in their region. (Marks: 3)**
3. Whose **average order value** is above the **company-wide average order value** . (Marks: 2)

Additionally, the company wants to:

4. Display **each such customer** along with: (Marks: 1)
    - Total spending in the last 12 months
    - Number of orders placed in the last 12 months
    - Average order value
    - Customer's country/region
5. Exclude customers who have **any cancelled orders** in the last 12 months. (Marks: 1)
6. Sort the results in **descending order of total spending**. (Marks: 1)
7. Only display customers whose **most recent order** was placed in the last 6 months. (Marks: 1)

Question 3: LLO-03 Advanced Concepts (Joins, Subqueries, & Views)

[Marks: 26 | Time: 55 Minutes]

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1. Find all **job titles** where the **AI Impact Level** is higher than the **average AI Impact Level** for the same industry.
(Marks: 3)
2. List **industries** where the **median salary** is higher than the **overall median salary across all industries** but only for jobs with **automation risk > 50%**. Use **nested subqueries with at least two levels**.
(Marks: 3)
3. For each **location**, find jobs where the **Projected Openings (2030)** are higher than the **average projected openings of all jobs in the same location**. Use **correlated subqueries**.
(Marks: 3)

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4. Find all jobs that appear in both **"Technology"** and **"Healthcare"** industries. Use **INTERSECT**.
(Marks: 3)
5. List **locations** where the **average AI Impact Level** is higher than 70% **AND** the **average automation risk** is above 60%. Use **INNER JOIN** between jobs and aggregated location stats.
(Marks: 3)
6. UNION the list of jobs with **remote work ratio > 50% in 2024** and jobs with **projected openings > 1000 in 2030**.
(Marks: 3)

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7. Create a view HighRiskJobs_View that joins **Job Title, Industry, AI Impact Level, Automation Risk**, then SELECT jobs with **AI Impact Level > 80**.
(Marks: 2)
8. Create a view Diversity_View that joins **Job Title, Location, Gender Diversity**, and SELECT locations where **average Gender Diversity > 50%**.
(Marks: 2)

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9. Create a view Salary_Impact_View that calculates the **difference between Median Salary and Industry Average Median Salary** using a subquery, then SELECT jobs where this difference is **positive**.
(Marks: 2)
10. Create a view ProjectedGrowth_View that calculates **Projected Openings / Job Openings (2024)** ratio per job using a subquery, then SELECT jobs where the ratio is **greater than the company-wide average ratio**.
(Marks: 2)